SANDOZ

TFM PROPOSAL

PriceErosion

Problem Statement

Biosimilars are complex drugs produced from living cells and widely used to treat serious diseases such as cancer or autoimmune disorders. However, due to intense market competition they can suffer from price erosion. Understanding **how fast and how much prices decline at SKU level** is critical for pharmaceutical companies.

Proposed Approach

In this thesis, we aim to develop an AI-based approach to **predict BIO price erosion per SKU**, combining accurate forecasting with explainable insights suitable for real-world pharmaceutical decision-making. The project will leverage historical, anonymized sales and pricing data at SKU level to:

- **Build predictive machine learning models** in Python to forecast future BIO prices and erosion patterns.
- **Identify key price drivers** influencing erosion in the different markets
- **Integrate a Generative AI component** using a Large Language Model (LLM) to translate model outputs into clear, natural-language explanations of forecasts.

This combined predictive–generative approach aims to deliver both robust forecasts and intuitive interpretability.

Technical tools

The project will utilize LLMs in combination with Python and relevant libraries to implement and deploy the models effectively.